FONDAZIONE links PASSION FOR INNOVATION

Politecnico di Torino

Art AI

Automated Artwork **Metadata Generation** using Multimodal LLM

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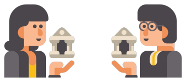
CHALLENGES

CONTEXT

Cultural institutions are experiencing a significant expansion in the digitization of their archives. Museums and art galleries, in particular, are working to convert extensive collections of artworks into digital formats that can be more easily preserved, searched, and accessed.

STAKEHOLDERS



Museum Curators

Manual annotation is:

- time-consuming;
- expensive;
- non-scalable.

AI Researchers

- Limited availability of high-quality labeled datasets;
- Lack of standardized metadata schemas across institutions.

CHALLENGES

Reduce the time and effort required for manual metadata compilation

Overcome the scarcity of reliable labeled data and the inconsistency of metadata structures across different cultural institutions

OBJECTIVES

Allowing curators to focus on **higher-value activities** through the automatic generation of metadata.

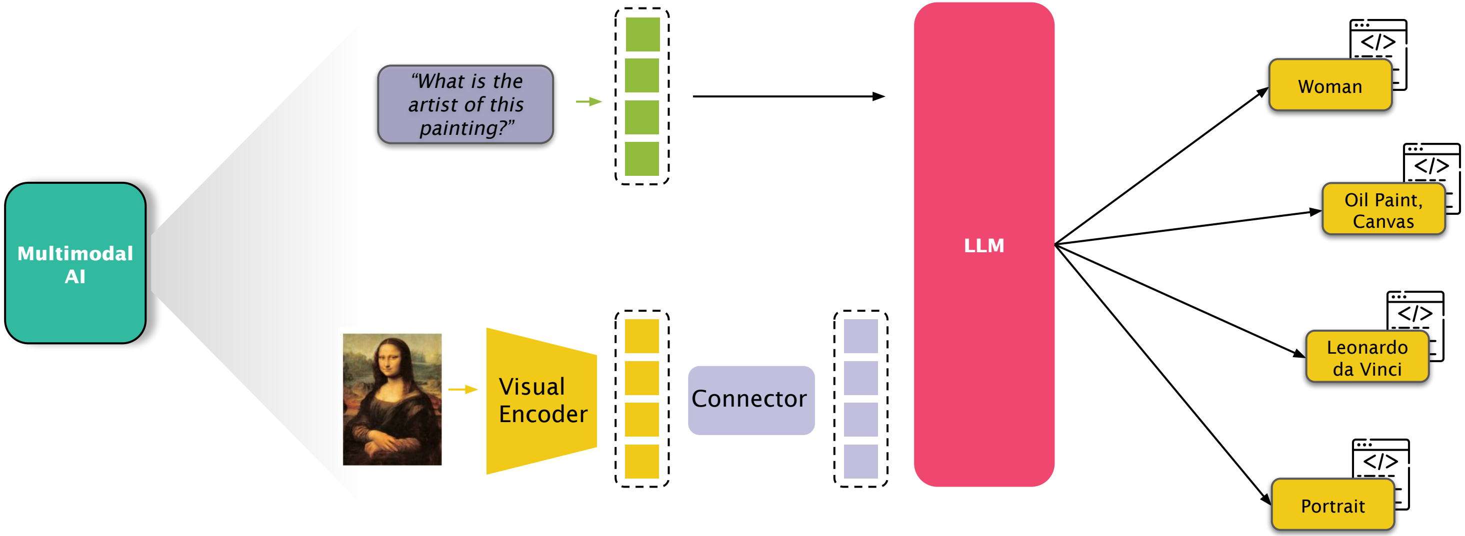
Enriching a database with more detailed information **improves searchability**, facilitates research, and enhances the understanding of the content.

Standardize data structures to enable easier **information sharing** and improve public access to cultural heritage.

HYPOTHESES

Q: Can a multimodal LLM correctly provide informations about a painting?

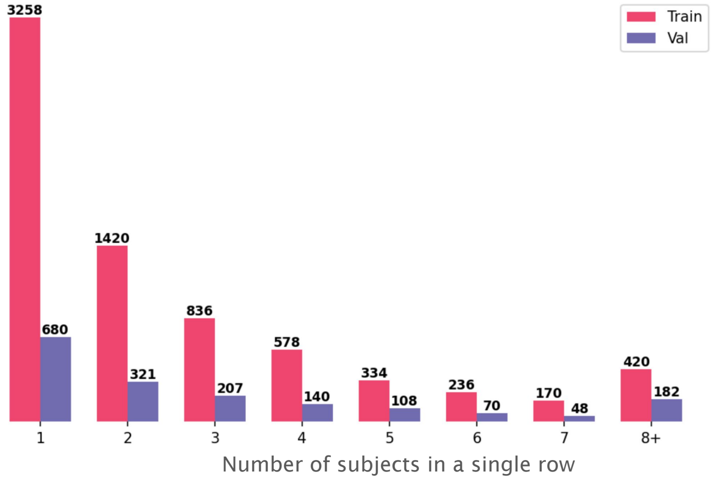
A: MLLM can effectively bridge the gap between raw visual data and structured knowledge



DATASET

artefact	type	title	description	genres	materials	subjects	authors	images
https:// wikidata	Painting	Madonna del Prato	painting by Raphael	['religio us art']	['oil paint', 'poplar panel']	['woman', 'boy', 'Mary', 'Christ Child',..]	['Raphael']	0299216 3-d6..

Column	Unique elements		Total occurrences		Average elements/row		Max elements/row	
	TRAIN	VAL	TRAIN	VAL	TRAIN	VAL	TRAIN	VAL
Authors	3.282	1.040	7.372	1.773	1,02	1,01	9	2
Genres	87	41	7.601	1.882	1,05	1,07	5	4
Materials	110	47	14.047	3.494	1,94	1,99	6	5
Subjects	6.217	2.442	20.497	6.446	2,83	3,67	69	76



DATASET

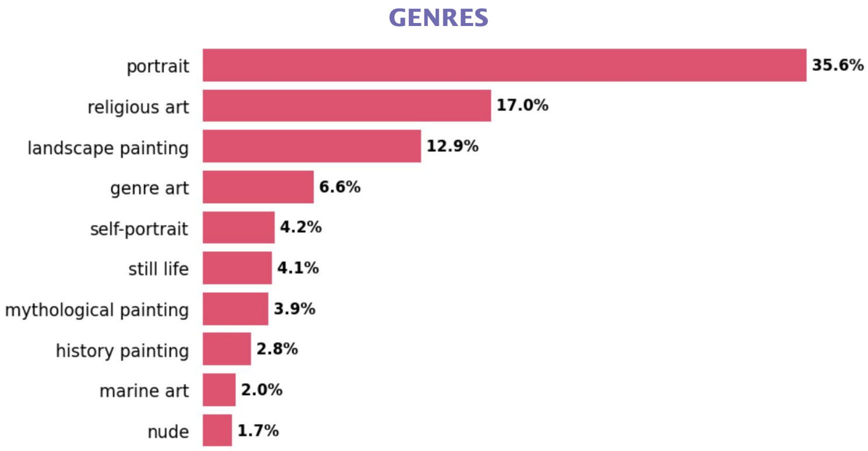
PREDOMINANT CLASSES

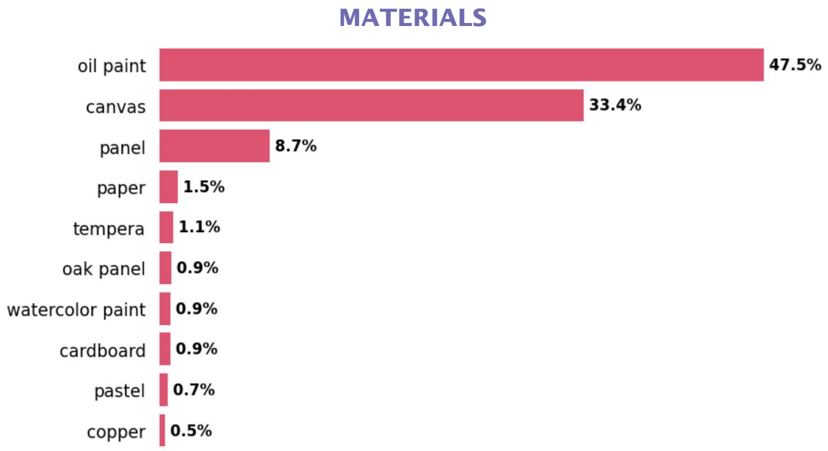
There is a clear dominance of a few labels compared to the entire vocabulary.

- In material, terms like "**oil paint**" or "**canvas**" cover a significant portion of the dataset.
- In genres, categories like "**portrait**" or "**religious art**" dominate over niche genres.

IMPACT ON MODELING

This imbalance introduces a risk of **bias**: the model may learn the dominant classes perfectly but fail to generalize on the rare ones (overfitting on dominant classes).





METHODOLOGY

ZERO–SHOT EVALUATION

Evaluating the models **"As-Is"** (Pre-trained weights) to benchmark their intrinsic capabilities without initial fine-tuning.

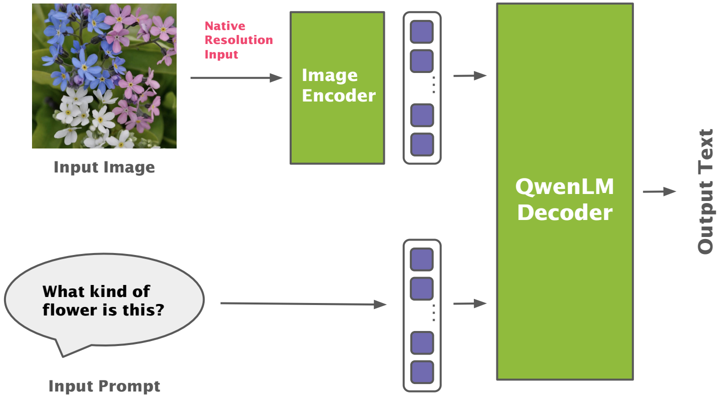
- **VLM SELECTION**
  We used **BLIP-2** and **Qwen2-VL** as our core Vision-Language Models to test varying architectures (Q-Former vs. Native Resolution).

FINE–TUNING

Fine-tune the model that demonstrates the highest baseline performance in benchmarks

- **PEFT TECHNIQUES**
  Updates only a small subset of parameters avoiding to retrain the entire model.

QWEN2-VL-2B-INS.

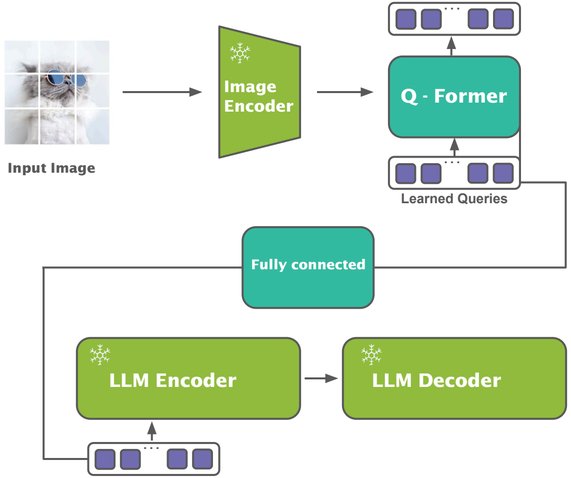


CHAT-TEMPLATE

NATIVE RESOLUTION

MULTIMODAL ROTARY POSITION EMBEDDING

BLIP-2 FLAN-T5-XL



INSTRUCTION-TUNED

FROZEN

INFORMATION BOTTLENECK

METHODOLOGY

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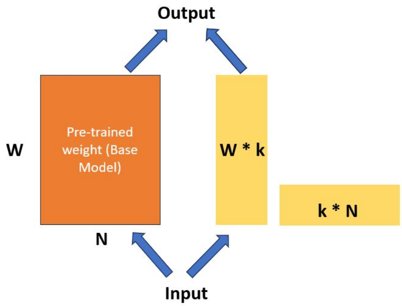
METHODOLOGY

PEFT (Parameter-Efficient Fine-Tuning)

Family of techniques that adapt large pre-trained models to new tasks by training only a small subset of parameters.

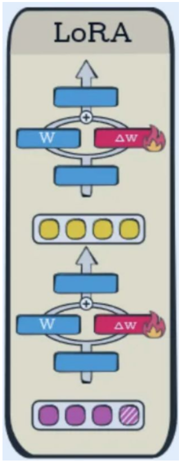
- **BENEFITS**
  Reduces memory usage, training time, and computational cost while preserving most of the model’s performance.



LoRA (Low-Rank Adaptation)

PEFT method that injects low-rank matrices into selected layers (typically attention layers).

- **DETAILS**
  Only low-rank parameters are trained, while the original model weights remain frozen.
- **OUR CONFIGURATIONS**
  - Rank = 8 – Dropout = 0.1
  - Rank = 16 – Dropout = 0.05



METHODOLOGY

LoRA TARGET MODULES

- **Vision-Language Adapter** (**visual.merger.mlp**)
  Updates the Multi-Layer Perceptron within the visual merger.
- **Self-Attention Layers** (**self.attn.**)
  Updates the Query, Key, Value and Output projections within the attention mechanism.
- **Feed-Forward Network** (**mlp.**)
  - Gate → filters the information
  - Up → maps inputs into a higher-dimensional space to extract complex features.
  - Down → compresses processed data back to the original space to merge with the network

```
lora_config = LoraConfig(
    r=16, lora_alpha=32,
    target_modules=[
    "visual.merger.mlp.0",
    "visual.merger.mlp.2",
    "model.layers.*.self_attn.q_proj",
    "model.layers.*.self_attn.k_proj",
    "model.layers.*.self_attn.v_proj",
    "model.layers.*.self_attn.o_proj",
    "model.layers.*.mlp.gate_proj",
    "model.layers.*.mlp.up_proj",
    "model.layers.*.mlp.down_proj"],
    lora_dropout=0.05, bias="none",
    task_type=TaskType.CAUSAL_LM)
```

LOSS FUNCTION

- **Cross Entropy** (**CAUSAL_LM**)
  - Standard Next-Token Prediction loss used to train LMs.
  - Measures the discrepancy between prediction and GT.
  - Penalizes incorrect token predictions in a sequence

pred:	["Subject:" ,	"man"]
gt:	["Subject:" ,	"woman"]
loss:	[**low** ,	**high**]

METHODOLOGY

SINGLE-MODEL

SINGLE PROMPT

A holistic approach where the model is asked to predict all four categories (Authors, Subjects, Materials, Genres) within a single inference cycle.

Pro: Comprehensive Image Analysis

Cons: Multitask Complexity

1IMG - 4 PROMPT

Four independent prompts are generated for each image, with each prompt focusing on a single metadata category (Authors, Subjects, Materials, or Genres).

Pro: Targeted Task Analysis

Cons: Computational Expensive (Dataset 4x)

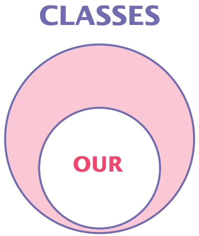
MULTI-MODEL

Four distinct versions of the model are fine-tuned independently, each dedicated to a single metadata category (Authors, Subjects, Materials, or Genres).

Pro: Task Specialization

Cons: High Management Complexity

Experimental setup



HYBRID TASK

- **QA problem:** VLM generate an answer as close as possible to the expected Wikidata answer.
- **Open-domain vocabulary:** the possible classes are not defined in advance.

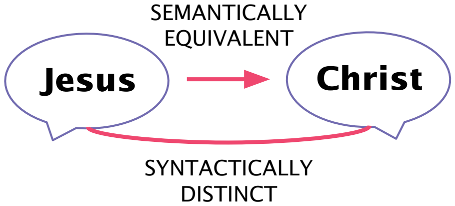
SYNTACTIC EVALUATION

How many terms are correctly predicted? We cannot use standard precision, recall, F1 → EXACT MATCH

SEMANTIC EVALUATION

Exact Match cannot capture semantic similarity → MPNET EMBEDDINGS

⚠ Similarity('person',any author) ≥ 0.7



QA METRICS

- BLEU and ROUGE useful for the N-gram overlap analysis.
- These metrics expect answers with more than one token. Particularly, BLEU penalizes answer with only one token.

EXACT MATCH EXTENSION

Querying Wikidata database to retrieve aliases of authors

WIKIDATA

RESULTS
ZERO-SHOT

	Exact Match (Accuracy)		Similarity Score		BLEU		ROUGE-1	
	QWEN	BLIP	QWEN	BLIP	QWEN	BLIP	QWEN	BLIP
Authors	**0.150**	0.001	**0.800**	0.684	X	X	**0.154**	0.003
Subjects	**0.014**	0.000	0.717	**0.722**	**0.030**	0.014	**0.193**	0.099
Materials	**0.218**	0.000	**0.886**	0.838	**0.210**	0.105	**0.609**	0.478
Genres	0.064	**0.353**	**0.586**	0.549	**0.079**	0.077	0.324	**0.485**

QWEN WIN

RESULTS
SINGLE-MODEL

1IMG-4PROMPT

Exact Match (Accuracy)

	Authors	Genres	Materials	Subjects
R=8, D=0.1	17,71%	75,80%	68,74%	**7,92%**
R=16, D=0.05	**18,45%**	**76,37%**	**69,19%**	7,74%

Exact Match (Accuracy)

	Scheduler	Authors	Genres	Materials	Subjects
R=16, D=0.05	✘	**18,45%**	**76,37%**	69,19%	7,74%
	✔	18,28%	75,68%	**69,59%**	**8,14%**

R=16, D=0.05 & SCHEDULER

SINGLE PROMPT

Exact Match (Accuracy)

	Authors	Genres	Materials	Subjects
R=8, D=0.1	**17,26%**	75,34%	66,57%	6,66%
R=16, D=0.05	17,06%	**75,57%**	**66,86%**	**7,00%**

Exact Match (Accuracy)

	Scheduler	Authors	Genres	Materials	Subjects
R=16, D=0.05	✘	17,06%	75,57%	66,86%	**7,00%**
	✔	**17,43%**	**75,97%**	**67,43%**	6,66%

R=16, D=0.05 & SCHEDULER

RESULTS
SINGLE-MODEL

1I

	Aut
R=8, D=0.1	17
R=16, D=0.05	18,

	Scheduler
R=16, D=0.05	✘
	✔

NICO

atch (Accuracy)

s	Materials	Subjects
%	66,57%	6,66%
%	**66,86%**	**7,00%**

t Match (Accuracy)

nres	Materials	Subjects
,57%	66,86%	**7,00%**
,97%	**67,43%**	6,66%

Exact Match (Accuracy)

	Strategy	Authors	Genres	Materials	Subjects
R=16, D=0.05 Scheduler	1IMG-4PROMPT	**18,28%**	75,68%	**69,59%**	**8,14%**
	PROMPT UNICO	17,43%	**75,97%**	67,43%	6,66%

1IMG-4PROMPT (R=16, D=0.05 & SCHEDULER)

YOU ARE THE BEST

R=16, D=0.05 & SCHEDULER

R=16, D=0.05 & SCHEDULER

RESULTS
MULTIMODEL

UPPER BOUNDS

The Multi-Model performance are used as our **upper bound**.

BENCHMARKING

Benchmark our **Best Configuration** (with Single-Model strategy) against this to validate its competitiveness and efficiency.

Exact Match (Accuracy)

	Authors	Genres	Materials	Subjects
MULTI-MODEL	**18,45%**	**76,65%**	**70,44%**	7,00%
BEST CONFIGURATION	18,28%	75,68%	69,59%	**8,14%**
	-0,17%	**-0,97%**	**-0,85%**	**+1,14%**

RESULTS
FINAL COMPARISON

BEST CONFIGURATION vs. ZERO-SHOT

A comparison between the best fine-tuned configuration and the zero-shot performance baseline **to quantify the performance gains**, considering all metrics.

	Exact Match (Accuracy)				Similarity			
	Authors	Genres	Materials	Subjects	Authors	Genres	Materials	Subjects
ZERO-SHOT BASELINE	15,00%	6,40%	21,80%	1,40%	80,00%	58,60%	88,60%	**71,70%**
BEST FINE-TUNED MODEL	**18,28%**	**75,68%**	**69,59%**	**8,14%**	**86,31%**	**75,68%**	**96,43%**	60,34%
	+3,28%	+69,28%	+47,79%	+6,26%	+6,31%	+17,08%	+7,83%	-11,36%

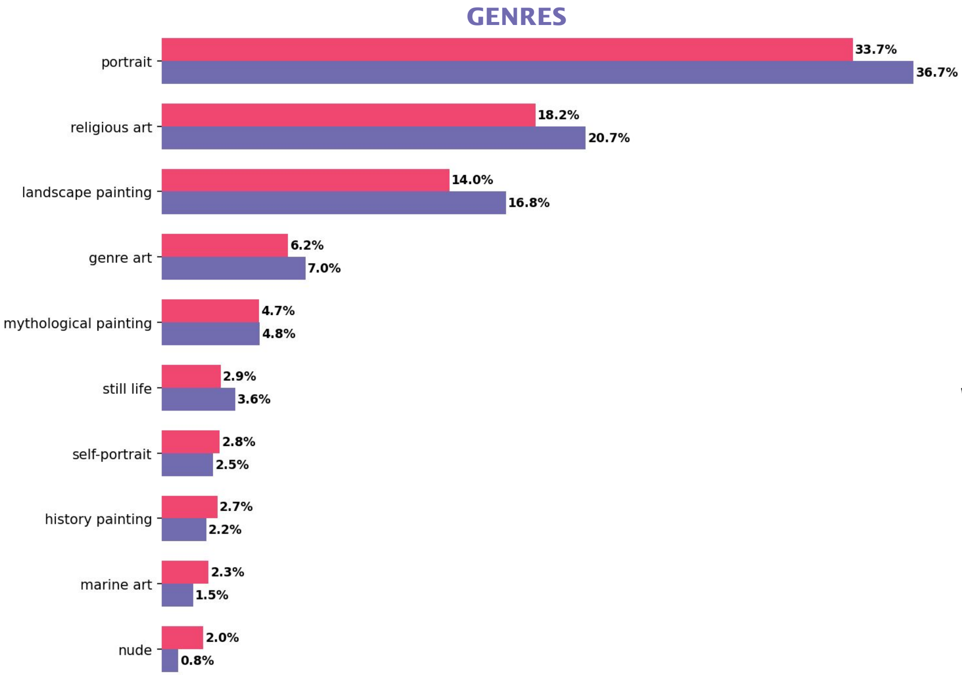
	BLEU				ROUGE-1			
	Authors	Genres	Materials	Subjects	Authors	Genres	Materials	Subjects
ZERO-SHOT BASELINE	x	7,90%	21,00%	3,00%	15,40%	32,40%	60,90%	19,30%
BEST FINE-TUNED MODEL	x	**20,07%**	**44,60%**	**4,63%**	**21,25%**	**80,56%**	**87,33%**	**24,99%**
	-	+12,17%	+23,60%	+1,63%	+6,15%	+48,16%	+26,43%	+5,69%

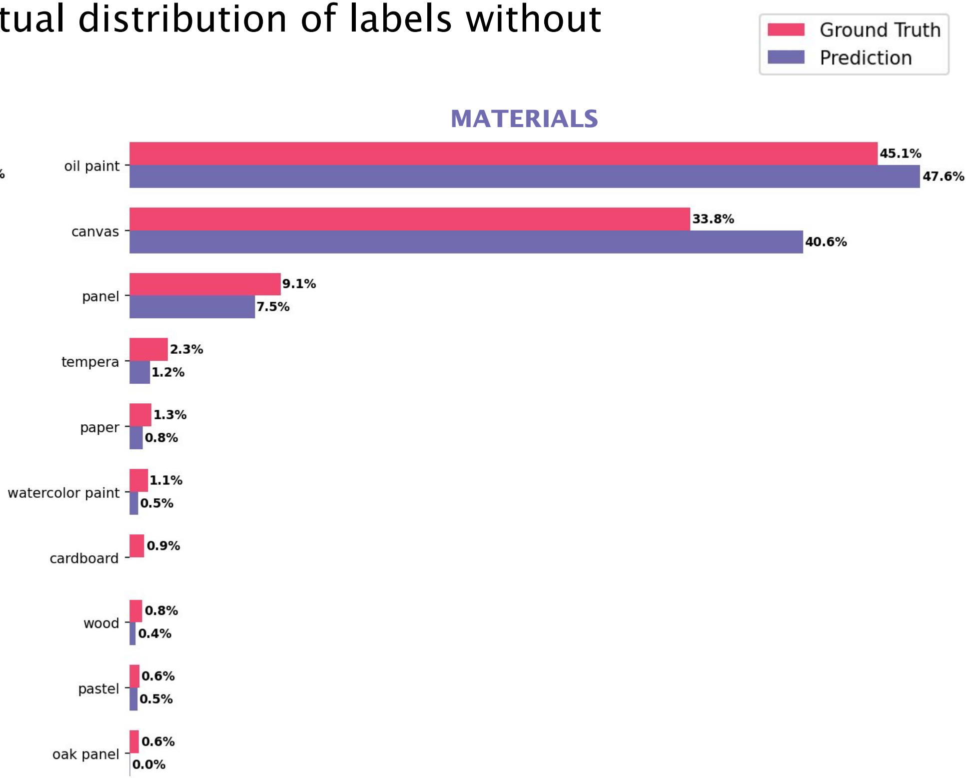
Mean percentage increase	
Exact Match	+31,65%
Similarity	+4,96%
BLEU	+12,47%
ROUGE-1	+21,61%

QUALITATIVE ASSESSMENT

IMPACT OF DATASET IMBALANCE ON PREDICTIONS

Despite the imbalance in our training data, the model demonstrates low bias and avoided overfitting. It accurately reflects the actual distribution of labels without excessively predicting the dominant classes.





CONCLUSIONS

- Approaches, like Parameter-Efficient Fine-Tuning (PEFT) and Low-Rank Adaptation (LoRA), improved model performance compared to the zero-shot baseline.

- Good result for “genres” and “material” columns, achieving an exact match accuracy of 69,28% and 47,79% respectively.

- Less performant results for “authors” and “subjects” columns:
    - **High Cardinality:** a vast amount of unique labels to predict.
    - **Lack of Standardization:** the level of detail is different, particularly for “subjects” column (detailed vs. generic).

POSSIBLE FUTURE DIRECTIONS

- Exploring alternative training strategies and loss functions
- Trying to create a standardized method for describing a work of art.

THANK YOU